

When the World Changes but the Score Does Not: A Counterfactual Audit of Computer-Use Agents

Vinh Duc Tran
Hanoi University of Science and Technology
vinh.td@hust.edu.vn

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Abstract

Computer-use agents are scored by whether an episode finishes and what a rubric judge assigns. That scalar mixes three questions: did the run terminate, did the answer reflect the world the agent actually saw, and did the judge notice when that world changed. We separate them with paired counterfactual episodes on MyPCBench. A locked intervention rewrites a pre-registered determining set D while the instruction, interface, and judge stay fixed. Tracking is read from the final answer against guest gold, not from the score. Score sensitivity is exact equality $S_{CF} = S_{Base}$ versus any nonzero delta.

On a frozen 5×10 design (50 intended cells; 4 never scheduled; 46 executed; 92 legs; 35 execution failures; 24 valid pairs), the three properties diverge. Claude: 9 valid pairs, 7 tracking-valid, 6 Type A, 1 score-sensitive, 2 Type B; invariance $6/7 = 0.857$ (Clopper–Pearson 95% CI [0.421, 0.996]). GPT-5.5: 8 valid, 7 tracking-valid, 3 Type A, 4 score-sensitive, 1 Type B; $3/7 = 0.429$ ([0.099, 0.816]). Qwen3.5-35B-A3B: one valid pair, Type A ([0.025, 1.000]). A designed trap becomes a positive control on one agent and opposite classes on another. Qwen3.5-9B (ablation) and Qwen3.8-Flash (exploratory) are not pooled into these rates.

A stable score does not certify tracking. A moving score does not certify a tracking failure. Completion, tracking, and score sensitivity are separable. Conventional CUA leaderboards do not run this check.

1 Introduction

Leaderboard evaluation of computer-use agents (CUAs) reports a completion score: the episode ended, and a rubric assigned a number [Jang et al., 2026, Xie et al., 2024]. That number is asked to stand in for three distinct facts. Did the harness record a terminal DONE? Did the agent’s answer match the environment state that was actually present? Did the judge’s score move when that state moved?

These facts need not agree. An agent can finish, miss a manipulated field, and still score 100 if the rubric does not pin that field. An agent can finish, report the manipulated field correctly on both sides of a paired comparison, and still see the score swing for reasons orthogonal to that field. Inferring either tracking or judge sensitivity from the scalar alone is therefore undefined.

This paper makes the three facts separately observable. For each frozen task we run a base episode in world E^0 and a counterfactual episode in $E^1 = I(E^0)$, where I is a locked SQL (and, when designed, file) patch on a pre-registered determining set D . Prompt, UI, screenshot size, and judge are held fixed. Guest probes confirm that gold moved. We code *completion* from the last

`traj.jsonl` action, *tracking* from the final answer against guest gold, and *score sensitivity* from exact scalar equality. Incomplete runs are execution failures, not tracking misses.

Contributions. (1) A three-way classification of valid pairs: Type A (tracked, score-invariant), score-sensitive (tracked, score moves), Type B (incomplete tracking, score high and unchanged). (2) Empirical separation on a frozen ten-task MyPCBench audit across three primary lanes, with a size ablation and an exploratory lane held out of the primary rates. (3) The observation that pre-registered “trap” labels do not determine observed class: one designed Type B cell is a clean positive control; another splits across agents.

The claim is methodological and local. We do not estimate a population rate of CUA failure, and we do not headline a pooled invariance fraction.

2 Setup

Intervention. For task t , E_t^0 is the seeded guest and $E_t^1 = I_t(E_t^0)$ is the same guest after a locked patch on D_t . Some Phase B tasks are dual-channel (sqlite and a paired local file must move together); others hold one channel constant as a distractor. The same agent A , instruction T_t , and judge J produce $\tau^0 = A(T_t, E_t^0)$ and $\tau^1 = A(T_t, E_t^1)$. Every intervention in this snapshot has a confirming guest probe; no pair is dropped for a failed inject.

What we read. $G(\tau)$ is the task-relevant quantities in the final answer (last assistant block in `messages.json` when present, else the `response` of the last `traj.jsonl` row). $S(\tau) = J(\tau) \in \{0, \dots, 100\}$. Writer-side `track` fields, `result.txt`, and the mere presence of `scores.json` are not used for DONE, tracking, or validity.

Completion. $\text{done}(\tau) = 1$ iff the last action is literally DONE. A pair is valid iff both legs are done.

Tracking. $\text{track} = 1$ iff $G(\tau^0)$ matches D_t on E_t^0 and $G(\tau^1)$ matches D_t on E_t^1 . Joint D_t requires every component on both legs. Partial credit is not used. Tracking is never inferred from S .

Score sensitivity. On a tracked pair, invariance is $S(\tau^1) = S(\tau^0)$ with no tolerance. Any nonzero delta is score-sensitive. Invariance is a property of J on this rubric, not a claim about model internals.

Classes (valid pairs only). **Type A:** $\text{track} = 1$ and $\Delta S = 0$. **Score-sensitive:** $\text{track} = 1$ and $\Delta S \neq 0$. **Type B:** $\text{track} = 0$ and the score remains high and unchanged relative to baseline (here: both scores ≥ 80 and $\Delta S = 0$ on the three primary Type B cells; ablation/exploratory Type B cells are 100→100). The unused cell of Figure 1 (miss + moving score) is not observed as a primary headline class in this snapshot.

Universe. Ten tasks from one pre-registered eligibility pool (seed 20260826): four Stage-4 locks and six Phase B additions (Table 1). Designed roles were frozen before runs. Observed class is coded after, against gold, not reverse-fitted to the design table.

Five lanes: Claude, GPT-5.5, and Qwen3.5-35B-A3B (primary); Qwen3.5-9B (size ablation); Qwen3.8-Flash (exploratory). Design matrix: 50 intended cells. Never scheduled, and

	score invariant	score moves
tracks D	Type A agent distinguishes E^0 and E^1 ; judge scalar does not	Score-sensitive agent distinguishes E^0 and E^1 ; judge scalar does
misses D	Type B incomplete tracking; score high and unchanged	not a class here (incomplete tracking with a moving score is reported as such if it occurs)

Figure 1: Taxonomy of valid pairs. Rows are agent behavior (final answer vs. gold). Columns are judge behavior (exact S). Type A is not automatically a rubric bug; Section 3.6 reads per-criterion arrays where they exist. Execution failures never occupy this grid.

Table 1: Frozen tasks and pre-registered design role. Observed class is reported in Section 3, not in this table.

Task	Phase	Designed role
retrieval-f001	A	control
aggregation-f003	A	control
preference_inference-f018	A	designed Type B (joint D)
counterfactual-f004	A	score-sensitive contrast
retrieval-f003	B	control
retrieval-f016	B	control
retrieval-f029	B	control (dual-channel)
retrieval-f030	B	designed Type B
aggregation-f018	B	designed Type B
preference_inference-f004	B	score-sensitive contrast

not execution failures: Qwen3.5-9B and Qwen3.8-Flash on **preference_inference-f018** and **counterfactual-f004** (4 cells, 8 legs). Executed: 46 cells, 92 legs. Of those legs, 57 are DONE and 35 are execution failures (15 of 35 on Qwen3.5-35B-A3B, mostly **EMPTY_XML**). Valid pairs: 24. Invariance rate for a model is Type A / tracking-valid (Type B excluded from that denominator). Intervals are exact Clopper–Pearson 95% CIs. Lanes are never pooled.

3 Results

The evidence that matters is that all three cells of Figure 1 occur, including on the same task across agents. Coverage tables follow the examples so that denominators are defined, not so that a rate can be quoted as the finding.

3.1 Type A: tracked, score unchanged

All five lanes complete **retrieval-f001** (loyalty tier and miles: Gold/38,450 → Silver/8,620) and all five track both legs. Scores stay at the base value (100 on four lanes; 80 on Qwen3.5-9B). The

rubric does not pin Gold or 38,450. Two correctly reported worlds can share a score; that is the eligibility design, not a hidden agent failure.

Claude **aggregation-f018** was *designed* as a Type B trap (sqlite charitable \$950 $\rightarrow$ \$0 and home-office 104 $\rightarrow$ 0 days; W-2/1099 and paired files held). Claude is the only lane that completed both legs (GPT’s CF leg is **EMPTY_XML**). Observed class is **Type A**: CF zeroes both sqlite fields, holds the untouched file figures, and flags the file/sqlite mismatch. Score 100 $\rightarrow$ 100. This is the cleanest positive control in the audit: joint *D* tracked, distractor channel correctly not moved.

Other primary Type A pairs (both legs match gold; $\Delta S = 0$): Claude **retrieval-f003** 65 $\rightarrow$ 65 (\$136,320 $\rightarrow$ \$80,000); **retrieval-f016** 100 $\rightarrow$ 100; **retrieval-f029** 100 $\rightarrow$ 100; **aggregation-f003** 80 $\rightarrow$ 80. GPT **retrieval-f001**, **retrieval-f003** 100 $\rightarrow$ 100, **aggregation-f003** 50 $\rightarrow$ 50. Qwen3.5-35B-A3B only on **retrieval-f001** 100 $\rightarrow$ 100.

3.2 Score-sensitive: tracked, score moves

preference_inference-f004 flips HangryDash’s top restaurant from Cooper’s Seafood House (88 orders) to Backyard Ale House (59), with TableFind held. Claude and GPT both report the flip and leave TableFind unchanged. Scores drop 100 $\rightarrow$ 79 (Claude) and 100 $\rightarrow$ 58 (GPT) because the rubric pins Cooper’s as HD top. Tracking succeeded; the judge responded. That is the opposite of Type B.

GPT additionally tracks three retrieval interventions whose scores move anyway: **retrieval-f016** 100 $\rightarrow$ 85 (cost basis \$8,213.25 $\rightarrow$ \$7,114.20, cash \$420 $\rightarrow$ \$50); **retrieval-f029** 33 $\rightarrow$ 100 (W-2 \$142k/\$28.4k $\rightarrow$ \$90k/\$18k; per-criterion arrays attribute the swing to items other than the wage figure); **retrieval-f030** 53 $\rightarrow$ 100 (TY2025 1099 held at \$1,200; charitable \$950 $\rightarrow$ \$100, file/sqlite mismatch flagged). Same agent, same protocol: Type A and score-sensitive co-exist.

3.3 Type B: incomplete tracking, score holds

Re-read against final DONE text and guest gold:

1. **Claude retrieval-f030**, 100 $\rightarrow$ 100. Designed move is sqlite charitable \$950 $\rightarrow$ \$100 with TY2025 1099 held at \$1,200. Base DONE answers *filed TY2024* 1099-NEC \$1,080, not the determining TY2025 \$1,200. CF reports \$1,200 and charitable \$100. Type B is the base-leg wrong year, not a CF miss of charitable.
2. **Claude preference_inference-f018**, 100 $\rightarrow$ 100. Joint *D*: GME 85 $\rightarrow$ 0 and OddsMarket YES 200/active $\rightarrow$ 0/settled. CF lists GME 0 shares / \$0 basis, and still lists GameStop YES as 200 shares / \$16.00 (same as base). One conjunct tracked; the joint set is not. Cleanest Type B.
3. **GPT counterfactual-f004**, 87 $\rightarrow$ 87. Guest zeroes the 1099 amount. CF still reports a \$1,200 stipend and \$1,068/year net. Not an execution failure; Claude’s base leg on this task is not DONE.

Ablation/exploratory Type B (not primary): Qwen3.5-9B **retrieval-f016** base reports \$8,788.75 vs. gold \$8,213.25 (CF exact \$7,114.20), scores 100 $\rightarrow$ 100; Qwen3.8-Flash **retrieval-f029** CF “Total Income \$91,200” vs. Box 1 gold \$90,000 (withholding \$18,000 exact), scores 100 $\rightarrow$ 100.

Table 2: Primary tier. Invariance = Type A / tracking-valid. CI is exact Clopper–Pearson 95%. $n \leq 9$; intervals are wide by construction.

Model	Valid	Tracking-valid	Type A	Sensitive	Type B	Invariance	95% CI
Claude	9	7	6	1	2	0.857	[0.421, 0.996]
GPT	8	7	3	4	1	0.429	[0.099, 0.816]
Qwen3.5-35B-A3B	1	1	1	0	0	1.000	[0.025, 1.000]

Table 3: Ablation and exploratory tiers. Not pooled with Table 2.

Model	Valid	Track.	A	Sens.	B	Inv.	95% CI
Qwen3.5-9B	3	2	1	1	1	0.500	[0.013, 0.987]
Qwen3.8-Flash	3	2	1	1	1	0.500	[0.013, 0.987]

3.4 Designed role vs. observed class

Pre-registration does not fix the outcome. **aggregation-f018** (designed Type B) → Claude Type A. **retrieval-f030** (designed Type B) → Claude Type B and GPT score-sensitive. **preference_inference-f018** (designed Type B) → Claude Type B as specified. **preference_inference-f004** (designed sensitive) → Claude and GPT sensitive. **counterfactual-f004** (designed sensitive) → GPT Type B: the intervention moved; the answer did not.

3.5 Per-model counts

Table 2 is the reportable unit. Qwen3.5-35B-A3B’s 1/1 is execution coverage (nine of ten cells fail), not an invariance finding.

3.6 Judge vs. agent: rubric arrays

On **aggregation-f003** (\$4,871.70 → \$400), four completing lanes track both totals. Claude and GPT are Type A; 9B and Flash are score-sensitive. GPT stays at [1,1,0,0] both legs: the refund-linked criteria pass twice; year-count/format criteria never fire. 9B moves [1,1,0,0] → [1,1,1,0]; Flash [1,1,1,0] → [1,1,1,1] — one extra criterion under CF, not a change in the reported total. Type A here is not “the refund criteria were blind.” Score-sensitive here is not “the agent failed to track.”

3.7 Cells outside this universe

hard_app-f033 and **situated_action-f028** are pilot-only and unaudited. No trajectory, guest file, patch, or writer output was modified to produce these numbers.

4 Discussion

Completion scores measure termination plus whatever the rubric binds. They do not, by themselves, measure whether D was read, and they do not, by themselves, measure whether J is attached to D . Type A shows the first without the second; score-sensitive shows both; Type B shows completion (and a high score) without tracking. That is the distinction this protocol is for.

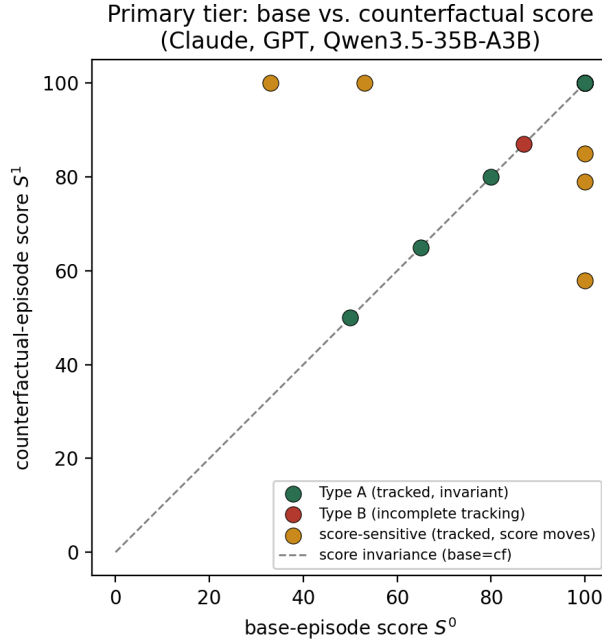


Figure 2: Primary valid pairs: S^0 vs. S^1 . Diagonal: invariant. Off-diagonal: score-sensitive. Type B points track incompletely despite the score.

Score invariance is not a defect to be patched out of every rubric. A channel-invariant item that asks whether the agent used the right app and reported a value can award the same score to two correct worlds (`retrieval-f001`). Changing that rubric would change the task. The audit’s job is to make the binding visible.

Designed traps are hypotheses, not labels. Reporting only the pre-registered role would have misstated Claude `aggregation-f018` and would have collapsed Claude and GPT on `retrieval-f030`.

5 Related work

OSWorld [Xie et al., 2024], WebArena [Zhou et al., 2024], and VisualWebArena [Koh et al., 2024] score agents with completion rubrics. MyPCBench [Jang et al., 2026] seeds a personal desktop that our I reuse. None of these leaderboards pair E^0 with $E^1 = I(E^0)$ to split tracking from S .

Shao et al. [2026] argue that a benchmark score supports a capability claim only when the intended capability remains necessary for success, and they audit shortcut paths. We audit a different cut: whether a gold-moving intervention that the guest confirms is reflected in the answer and in S . Complementary, not equivalent. Turk [2026] pair clinical cases with controlled mutations and find that coverage-style scores can hide update failures; we observe the same *pattern* (score vs. state change) for GUI CUAs on personal-data tasks. Bellibatlu et al. [2026] measure judge instability under prompt paraphrase; we measure judge (in)sensitivity to a verified world-state change, using the rubric’s own item array when available.

The pairing $E^1 = I(E^0)$ follows the interventionist language of Pearl [2009]. We do not identify internal causal structure of the model.

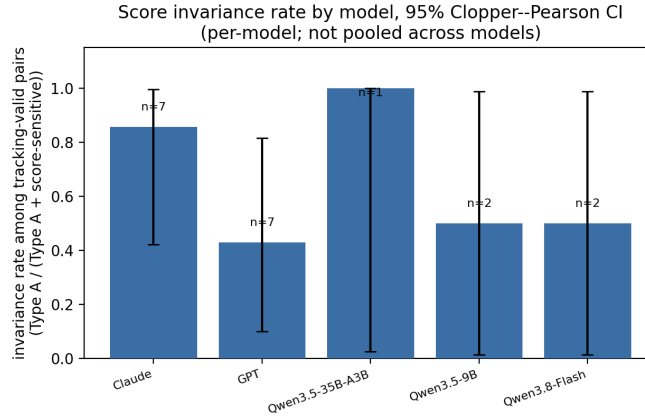


Figure 3: Per-model invariance among tracking-valid pairs. Bars are not a pooled rate.

6 Limitations

Ten tasks, one environment, five lanes, $n \leq 9$ valid pairs per primary model. CIs are wide; they bound binomial uncertainty on this sample, not a deployment rate. Qwen3.5-35B-A3B contributes one semantic pair. Classification of all 24 valid pairs is hand-coded from final-answer text against guest gold (`tracking_evidence.md`); traceable, not automated. Score deltas are attributed to rubric items only where `per_rubric_max` exists. Interventions test evaluator and agent behavior under controlled state changes, not arbitrary real-world shifts.

7 Conclusion

On this frozen audit, task completion, counterfactual tracking, and benchmark score sensitivity come apart. A stable score does not certify tracking; a moving score does not certify a tracking failure. Paired, gold-moving checks make that split observable. CUA leaderboards currently do not run them.

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